

Level	BE	Full Marks	80
Programme	BEI	Pass Marks	32
Year / Part	III / I	Time	3 hrs.

Subject: - Operating Systems (CT 612)

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- How does an operating system act as an extended machine? Explain with an example. What are the design tradeoffs between monolithic and microkernel architecture? [4+4]
- Explain the significance of the Process Control Block (PCB). For the given information of the processes, find average TAT, average WT, CPU utilization, and throughput according to FCFS and SRTN algorithms. [2+8]

Process	Arrival Time	Burst Time	Priority
P ₀	1	2	1
P ₁	2	4	0
P ₂	0	10	3
P ₃	3	8	2

- What is race condition? How do you solve producer-consumer problem using Semaphore? Explain with pseudo-code. [1+6]
- Differentiate internal fragmentation and external fragmentation. How many page-fault for LRU, FIFO and Optimal page replacement algorithms with the following given reference string for three-page frames: 7,0,1,2,0,3,0,4,2,3,0,3. [3+6]
- Describe the contiguous and linked list methods for file allocation. Explain the concept of inodes and their role in UNIX/LINUX File System. [4+4]
- What are four necessary conditions of deadlock? Consider the following snapshot of the system: [4+6]

	Allocation				Maximum Claim				Available			
	A	B	C	D	A	B	C	D	A	B	C	D
P0	0	0	1	2	0	0	1	2	1	5	2	0
P1	1	0	0	0	1	7	5	0				
P2	1	3	5	4	2	3	5	6				
P3	0	6	3	2	0	6	5	2				
P4	0	0	1	4	0	6	5	6				

Is the system safe? If so, show the safe sequence.

- Explain goals of I/O software. A disk drive has 50 tracks, numbered 0 to 49. The read/write head is currently serving a request at track 15 and it was at track 48 previously, and the queue of pending requests, in FIFO order is 4, 40, 11, 35, 7, 14. What is the total track movement for head for the following algorithms: a) FCFS b) SSTF c) C-SCAN. [3+6]

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- Describe the functions of Operating System (OS) with examples. Differentiate monolithic and microkernel structure of OS. [5+3]
- Explain different states of a process. What are the differences between preemptive algorithms and non-preemptive algorithms for scheduling of a processes? Consider the following set of processes. [3+2+5]

Process	Arrival Time	Burst Time
A	0	3
B	1	5
C	3	2
D	9	5
E	12	5

Draw Gantt chart and calculate average weighting time for each process using following algorithms:

- Shortest Remaining Time First
 - HRRN
- What is counting semaphores? How do you solve consumer producer problem using semaphore? Explain with pseudo-code. [1+6]
 - Define logical and physical address. What is the importance of virtual memory? Consider the following page reference string with 3-page frames: [2+2+6]
1,2,3,4,2,1,5,6,2,1,2,3,7,6,3,2,1,2,3,6
How many page faults will occur for FIFO, LRU, and Optimal Algorithm?
 - Explain files allocation techniques with its advantages and disadvantages? Briefly explain the directory and its hierarchy. [4+4]
 - Explain the importance of device driver in I/O system. Suppose a disk has tracks numbered from 0-180. The disk queue has the following request tracks: 87, 170, 40, 150, 36, 72, 66, and 15. Suppose the current head is at track number 60, and it has just finished the request at 54. Find the total head movement for SSTF, SCAN, and C-LOOK algorithms. [2+6]
 - Differentiate between Encryption and Decryption and explain their role in enhancing security of the system. What do you mean by Access Control List (ACL)? [4+2]
 - What are the technical and soft skills that a system administrator must acquire? Explain tasks performed for user account management. [4+2]

9. What would be necessary steps to prevent deadlock? Considering a system with five processes P₀ through P₄ and three resources of type A, B, C. Resource type A has 10 instances, B has 5 instances and type C has 7 instances. Suppose at time t₀ following snapshot of the system has been taken.

[2+2+4]

Process	Allocation			Max need		
	A	B	C	A	B	C
P ₀	0	1	0	7	5	3
P ₁	2	0	0	3	2	2
P ₂	3	0	2	9	0	2
P ₃	2	1	1	2	2	2
P ₄	0	0	2	4	3	3

- Calculate Need matrix
 - Is the state safe to run all the processes? If so, show the safe sequence of process execution.
10. Write short notes on: (Any Three)
- Shell Programming
 - Critical Section
 - TSL Instruction
 - Thrashing
 - Memory Compaction
 - Thrashing

[3×3]

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1. What is an operating system? Explain how operating system acts as a resource manager. Why operating system is termed as virtual machine? Explain. [1+4+4]
2. What do you mean by preemptive and non-preemptive scheduling? Schedule the following processes according to SRTN and priority preemptive algorithm to calculate average TAT and WT. Here, the process with the lowest value has the highest priority. [2+8]

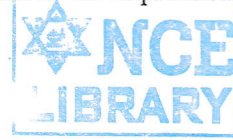
Process	Arrival Time(ms)	Burst Time(ms)	Priority
P0	0	8	2
P1	2	2	4
P2	4	12	0
P3	8	4	3
P4	12	10	1

3. What is semaphore? Solve reader-writer problem using semaphore. [6]
4. Consider the following page reference strings; 2,3,4,2,1,3,7,5,4,3,1,5. Find how many pages fault occur according to OPTIMAL, LRU (least recently used) and LFU (least frequently used) page replacement algorithm assuming 3-page frames. [6]
5. What is file attribute? Explain all the file allocation methods with its advantages, disadvantages and suitable block diagram. Explain how a file system performance can be increased. [2+6+2]
6. Explain the working of the DMA Controller with the respective figure. Consider a disk queue with requests for I/O to blocks on cylinders 175, 155, 188, 23, 169, 15, 130 for cylinders from 0-199 (200 cylinders). The current head position is at 100 which had just finished the request at cylinder 109. Starting from the current head position, find total head movement for FCFS, SSTF, and SCAN disk arm scheduling algorithms. [4+6]
7. What are the four essential conditions for deadlock? How deadlock is recovered? Explain. A system has 2 process and 3 resources. Each process need maximum of two resources, is deadlock possible? [4+3+3]
8. Explain about Protection Domain and Access Control List (ACL). Why 'HASH' function is called Message Digestor? Explain. [2+2+2]
9. Define system administrator. What are the different technical and soft skills that one should pursue as a system administrator? [1+6]
10. Write short notes on: (Any Two) [2×3]
 - a) Principal of IO software
 - b) Compaction
 - c) Ceaser cipher

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1. How does operating system provide a beautiful interface to user? Why Exo-Kernel doesn't require Re-mapping of resources? Is layered structure of operating system is better than monolithic structure? Explain. [3+3+3]
2. Why threads are called light weight process? Consider the following set of processes, with length of the CPU burst time given in milliseconds. [2+8]

Process	Burst time	Priority
P1	0	3
P2	2	6
P3	4	4
P4	6	5
P5	8	2

All the processes are assumed to have arrived in order all at time 0.

- (i) Draw Gantt Chart Using FCFS, SJF scheduling algorithm.
- (ii) Find average turnaround time and waiting time for each scheduling algorithm.
3. Explain about Lock variable for achieving Mutual Exclusion. Explain the types of semaphore along with the major operations of semaphore with a simple Pseudocode. [2+5]
4. Explain how a logical address is mapped to a physical address in paging. How is virtual memory management done? [4+2]
5. How many pages fault for the following given reference string for four-page frames 0,9,0,1,8,1,8,7,8,7,1,2,8,2,7,8,2,2,8,3. [7]
 - a) LRU
 - b) FIFO
 - c) Optimal page
6. Explain I-node approach of files implementation with its advantages and disadvantages. [5]
7. Consider a disk drive having 100 cylinders. The head is currently serving request at cylinder 43 and the previous request was at 56. The queue of pending request is 86,70,13,74,48,9,22,50,30. Starting from current head position, what is total head movement (in cylinders) to service the pending requests of each of following disk arm scheduling algorithms. [6]
 - a) FCFS
 - b) SSTF
 - c) SCAN
 - d) LOOK

8. Explain the resource allocation graph with an example. Consider a system with 4 concurrent processes (P1, P2, P3, P4) and three resource types A, B, and C with total instances 7, 10, and 10 respectively. The allocation and maximum claim at state t_0 is: [2+8]

Process	Allocation			Maximum Claim		
	A	B	C	A	B	C
P1	1	2	0	3	5	0
P2	3	0	2	5	0	5
P3	1	2	2	2	2	5
P4	0	4	0	0	10	2

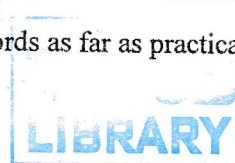
- What is the total number of available resources?
 - What will be the need matrix?
 - Is the system safe? If so, show the safe sequence.
9. What do you mean by ACL? How is it different from the capabilities list? How does Caesar Cipher convert plain text to ciphertext? [2+2+4]
10. What are the roles of a system administrator? How is a special user different from a general user? Explain. [4+2]
11. Write short notes on: (Any Two) [2×3]
- Compaction
 - Thrashing
 - Belady's anomaly

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1. Why operating system evolve over long periods of time? Justify how OS act as resource manager. [1+3]
2. Explain fork() and spawn() system calls in the OS. Schedule the following set of process according to Round-Robin scheduling a algorithm with Quantum time = 4 ms and calculate the average waiting time and average Turn-around time, throughput and CPV utilization. [3+5]

Process	Arrival time (ms)	CPU time (ms)
A	0	12
B	2	8
C	5	7
D	10	9

3. Why do we organize disk as RAID? Suppose a disk drive with 150 cylinder numbered from 0-149. The drive is currently serving a request at 35 and the previous request is at disk 55. The queue of pending request is 98, 103, 38, 122, 10, 128, 65, 75. Starting from the current head position, Calculate the total head moment (in cylinder) that the disk arm moves to satisfy all the pending request for SSF, SCAN, and LOOK Disk Arm Scheduling Algorithm. [2+6]
4. a) Explain first fit, Next fit memory allocation algorithm with an example. [5]
- b) How semaphore is used in process synchronization? Explain how semaphore is best solution for producer consumer problem with pseudo-code of both producer and consumer process. [1+7]
5. What is file attribute? List the file system performance indicator. Explain the file allocation methods with its advantages and disadvantages. [1+2+5]
6. What are the differences between fixed partitioning and variable partitioning system of memory for multiprogramming? Given reference to the following pages by a program:
0,9,0,1,8,1,8,7,8,7,1,2,8,2,7,8,2,3,8,3
How many page faults will occur if the program has 4 frames for Optimal Page Replacement Algorithm? [3+5]

7. Explain necessary conditions for deadlock. Consider the following snapshot of a system (P = Process, R = Resource): [2+6]

Available

RA	RB	RC	RD
8	5	9	7

Maximum Demand

	RA	RB	RC	RD
P0	3	2	1	4
P1	0	2	5	2
P2	5	1	0	5
P3	1	5	3	0
P4	3	0	3	3

Current Allocation

	RA	RB	RC	RD
P0	1	0	1	1
P1	0	1	2	1
P2	4	0	0	3
P3	1	2	1	0
P4	1	0	3	0

By using Bankers algorithm, Calculate the need matrix. Is the system safe? If safe find safe order of process.

8. Describe the role and responsibilities of system administrator to keep the system updated and efficient. Explain with examples. [6]
9. Define ACL. Write the type of security breach in following attack case? Also suggest a solution in each to prevent the attack. [2+2+2+2]

"Ramesh found that Nirmal's Facebook was login in Computer Lab. He then changed the personal information and login credentials of Nirmal's account."

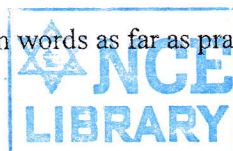
10. Write short notes on: [3×3]
- Ostrich algorithm
 - Shell programming
 - Process control Block

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1. What is an Operating system? Explain the statement “ operating system acts as a Broker between hardware and application program”. [5]
2. What is priority of a process? Why do we need it? Explain. Make a schedule for the processes mentioned in the table below as per Shortest Remaining Time First (SRTF) algorithm. Also calculate average turnaround time and average waiting time, throughput and CPU utilization. [2+6]

Process	Burst Time	Arrival Time	Priority
P ₁	3	0	3
P ₂	6	1	5
P ₃	1	2	2
P ₄	4	3	1
P ₅	2	4	4

3. What are the principles of I/O software? Suppose that a disk drive has 100 cylinders, numbered 0 to 99. The drive is currently serving a request at cylinder 43 and previous request was at cylinder 25. The queue of pending request, in FIFO order is: 86,70,13,74,48,9,22,50,30. Produce the schedules to satisfy all the pending requests for each of the following disk scheduling algorithms? [2+6]
 - a) FCFS
 - b) C-SCAN
4. a) Explain Best fit and Worst fit memory allocation algorithm with an example. [5]
 - b) What is the need process synchronization? How can the semaphore solve the reader-writer problem? Explain with respective pseudo-code of both reader and writer process. [2+6]
5. What is directory organization in files? Explain its types. [8]
6. What is Belady's anomaly in FIFO? Consider the following page reference strings; 2, 3,4,2,1,3,5,4,3,1,5,3,4,5,0,1,4,2. Find how many page fault occur according to OPTIMAL, LRU (least recently used) and LFU (least frequently used) page replacement algorithm assuming 3 page frames. [1+7]

7. Consider a system with 5 concurrent processes (P0, P1, P2, P3, P4) and 4 resource types (R0, R1, R2, R3). The number of instances of each resource type in the system are 6,4,4,2 respectively. Allocation table and Maximum claim table are as follows: [8]

Maximum claim

	R0	R1	R2	R3
P0	2	0	1	1
P1	1	1	0	0
P2	1	1	0	0
P3	1	0	1	0
P4	0	1	0	1

Allocation

	R0	R1	R2	R3
P0	3	2	1	1
P1	1	2	0	2
P2	1	1	2	0
P3	3	2	1	0
P4	2	1	0	1

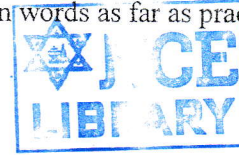
- Calculate Need matrix as per Banker's algorithm.
 - Is the state safe? If so, show the safe execution of the processes.
8. What are the roles and responsibilities of system administrator? [6]
9. Explain the types of Network Attacks? What is ACL? Why 'HASH' function is called Message Digestor? [2+2+2]
10. Write short notes on: [3×3]
- Thrashing
 - Virtual Machine
 - User level thread vs Kernel- level thread.

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1. a) Explain OS as an Extended Machine. Differentiate between Monolithic Kernel and Micro-Kernel. [2+4]
b) Explain operating system as a virtual Machine. [4]
2. a) Differentiate between Preemptive and Non-Preemptive Scheduling. Apply MLQ scheduling for following set of processes of two queues Q1 and Q2 where Priority of Q1 is greater than that of Q2 and Q1 uses Round Robin (Time Quantum = 2) and Q2 uses FCFS.

Process	AT	BT	Queue Number
P1	0	5	1
P2	2	12	2
P3	5	3	1
P4	10	6	1

- Construct Gantt –Chart and compute average TAT for above scenario. [2+4]
- b) What is multithreading? Explain five state process model with figure. [4]
 3. a) Why process need to be synchronized? Explain Peterson's Solution in mutual exclusion. [2+3]
b) What is Semaphore? How can Producer Consumer problem be solved using Semaphore? Explain. [1+4]
 4. a) Why multilevel paging is required? [2]
b) Consider the following page reference string: 5,0,2,1,0,3,0,2,4,3,0,3,2,1,3,0,1,5
Calculate page hit percentage. How many page faults would occur for the FIFO, Optimal, LFU and LRU replacement algorithms having four frames? Remember all frames are initially empty, so your first unique page will cost one fault each. [8]
 5. What are the different methods for allocating disk space for file? Explain free space management techniques. [3+7]
 6. What are the functions of device independent I/O software? Suppose that a disk has 5000 cylinders, numbered from 0 to 4999. The drive is currently serving a request at cylinder 143, and the previous request was at cylinder 125. The queue of pending requests in FIFO order is:

86,1470,913,1774,948,1509,1022,1750,130

Starting from the current head position, what is the total distance (in cylinders) that the disk arm moves to satisfy all the pending requests, for FCFS, SSTF, C-SCAN and C-LOOK Algorithm? [2+8]

7. Explain Coffman conditions for deadlock. Consider the following snapshot

Processes	Allocation			Max			Available		
	A	B	C	A	B	C	A	B	C
P0	0	1	0	7	5	3	3	3	2
P1	2	0	0	3	2	2			
P2	3	0	2	9	0	2			
P3	2	1	1	2	2	2			
P4	0	0	2	4	3	3			

Is the state safe? If so, show the safe execution of processes.

[3+7]

8. Write short notes on:

[4×2.5]

- Cryptography
- Access Control List
- Roles of System Admin
- AWK Tool

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1. a) Why is the process table needed in a timesharing system? Is it also needed in personal computer systems running UNIX or Windows with a single user? [6]
b) Distinguish between Shell and Kernel. [4]
2. a) What are the advantages and disadvantages of implementing threads in user space? [4]
b) Let us consider five processes with given arrival time and length of the CPU burst given in milliseconds. [6]

Process	Arrival time	CPU time
P1	0	9
P2	1	5
P3	2	2
P4	3	6
P5	4	8

Calculate the turnaround time and waiting time for all processes applying First Come First Serve, Shortest Job first and Round Robin (time quantum = 3) algorithms.

3. Define race condition. What are the requirements of mutual exclusion? How can you achieve mutual exclusion using Peterson's Solution? Explain with pseudo code. [2+2+6]
4. a) Differentiate between Compaction and Coalescing technique. [4]
b) Consider a swapping system in which memory consists of the following hole sizes in memory order: 10 MB, 4 MB, 20 MB, 18 MB, 7 MB, 9 MB, 12 MB and 15 MB. Which hole is taken for successive segment requests of [6]
(i) 12 MB
(ii) 10 MB
(iii) 9 MB
for first fit? Now repeat the question for best fit and worse fit.
5. a) Explain various ways of implementing file system. [6]
b) How do you measure the file system performance and how it can be improved? [4]
6. What do you mean by RAID? Suppose a disk with 200 cylinders numbered from 0-199. The drive is currently serving a request at 45 and previous request was at 125. The queue of pending request is 105, 178, 23, 67, 43, 78, 167, 56 and 98. Starting from current head position, calculate the total head movement (in cylinder) that the disk arm moves to satisfy all pending request for SSTF, SCAN, LOOK and C-SCAN disk scheduling algorithm. [2+8]

7. Explain in detail how can detect deadlock in operating system. Consider the following system with resources A, B, C, D and process P0 to P4. Is the state safe? If so, show the safe execution of processes.

[3+7]

Process	Max				Allocation				Available			
	A	B	C	D	A	B	C	D	A	B	C	D
P0	6	0	1	2	4	0	0	1	3	2	1	1
P1	1	7	5	0	1	1	0	0				
P2	2	3	5	6	1	2	5	4				
P3	1	6	5	3	0	6	3	3				
P4	1	6	5	6	0	2	1	2				

8. Write short notes on Caesar Cipher and Access Control Lists.
9. What is the significance of system Administration? Describe the roles and responsibilities of system administrator in Insurance Company.

[3+3]

[4]
